



Lecture 6

- Gauss's law
- In-class practice

In-class problems 2

[2026 In-Class Problems 2.pdf](#) ↓

2026 In-Class Problems 2.pdf



Review: Calculate the electric field from different charge distributions using Coulomb's law

Steps of using Coulomb's law to calculate $\vec{E}$ field

1. Choose coordinate system. define origin
2. Write radius vector of the source (charge) $\vec{R}_0$
3. Write radius vector of point of observation $P = \vec{R}$
4. Find vector from source to point $\vec{R} - \vec{R}_0$
5. Apply Coulomb's law
6. Sum or Integrate to find $\vec{E}$ at point of observation P .
 $\downarrow$ point source $\downarrow$ charge distribution

$\Rightarrow$ Volume ... (ρ_v)
 $dv \Rightarrow dQ = \rho_v dv$

$$d\vec{E} = \frac{1}{4\pi\epsilon} \frac{dQ}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$

$\Rightarrow$ point charges $\frac{Q_1}{E_1}, \frac{Q_2}{E_2}, \dots, \frac{Q_N}{E_N}$
 $\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_N$

$\Rightarrow$ line charge distribute ($\rho_L = \frac{dQ}{dl}$)
 $dl \Rightarrow (dQ = \rho_L dl)$ $\vec{E} = \int_{\text{line}} \frac{1}{4\pi\epsilon} \frac{\rho_L dl}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$

$\Rightarrow$ surface ... ($\rho_s = \frac{dQ}{ds}$)
 $ds \Rightarrow dQ = \rho_s ds$
 $\vec{E} = \int_s d\vec{E} = \int_{\text{surface}} \frac{1}{4\pi\epsilon} \frac{\rho_s ds}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$

Gauss's law



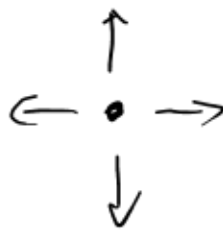
Differential form :
(Point-based)

electric flux density $\left[\frac{C}{m^2}\right]$

$$\nabla \cdot \vec{D} = \rho_v$$

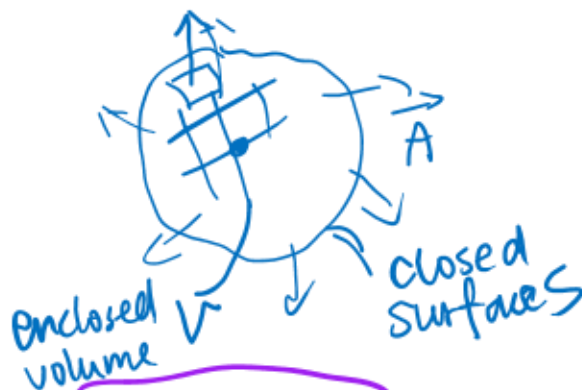
volume charge density $\left[\frac{C}{m^3}\right]$

$$\vec{D} = \epsilon \vec{E}$$



Divergence Theorem

Integral form



$$\oint_S \vec{A} \cdot d\vec{S} = \int_V \nabla \cdot \vec{A} dv$$

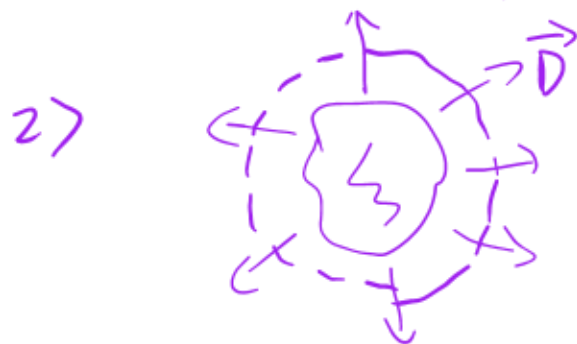
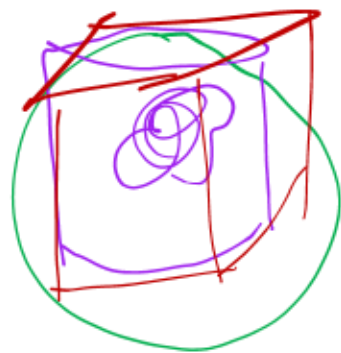
flux of $\vec{A}$
thru the closed
surface

$$\oint \vec{D} \cdot d\vec{S} = \int_V \nabla \cdot \vec{D} dv = \int_V \rho_v dv = Q_{\text{enclosed}}$$

Q_{enc}
total charge enclosed by the
surface S.

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{enc} = \int_V \rho_v dv$$

Notes: 1) closed surface can be any arbitrary surface.



$$\vec{D} \cdot d\vec{S} > 0 \quad Q_{enc} > 0 \quad \text{Source}$$



$$\vec{D} \cdot d\vec{S} < 0 \quad Q_{enc} < 0 \quad \text{Sink}$$

3) Although integral form of Gauss's law is universally true, it's not always useful to compute $\vec{E}$

Unless the charge distribution possess symmetry properties, which allows you to find a surface over which $\vec{E}$ ($\vec{D}$) is constant in magnitude, and its direction is normal/tangential to the surface.

Gauss's law

Steps:

1. Check the symmetry of the charge – **critical**

Make sure the charge distribution has sufficient symmetry so that you know the **direction of D** and can find a surface on which its **magnitude is constant**.

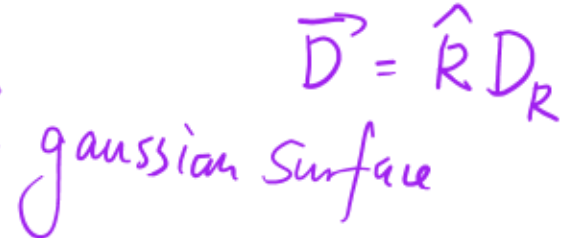
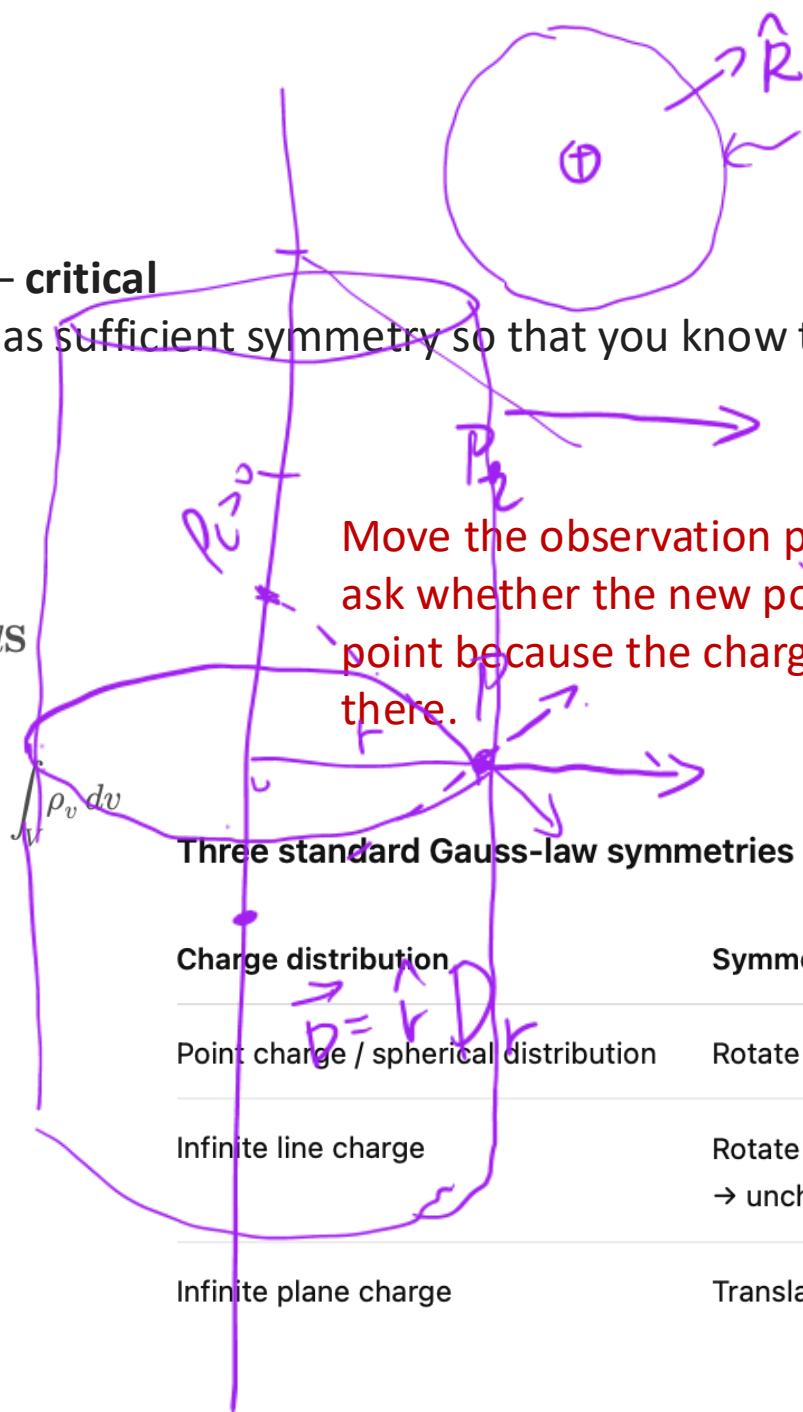
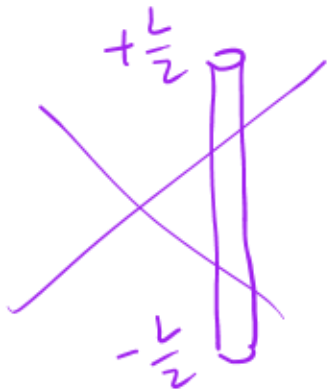
2. Choose that Gaussian surface

3. Evaluate the left-hand side: $\oint_S \mathbf{D} \cdot d\mathbf{S}$

4. Evaluate the right-hand side: $Q_{\text{enc}} = \int_V \rho_v dv$

5. Find **D, or E**

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon}$$



Move the observation point using that rotation/translation, and ask whether the new point is physically equivalent to the original point because the charge distribution looks exactly the same from there.

Three standard Gauss-law symmetries

Charge distribution

Point charge / spherical distribution

Infinite line charge

Infinite plane charge

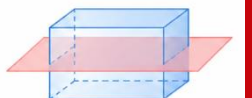
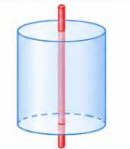
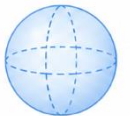
Symmetry test

Rotate about the center → unchanged

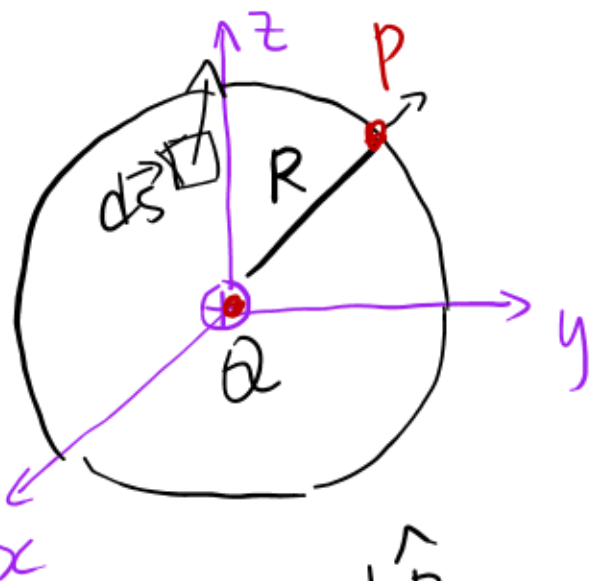
Rotate around line **and translate along line** → unchanged

Translate anywhere parallel to plane → unchanged

Example Gaussian surface



Gauss's law



$$\vec{D} = \hat{R} D_R$$

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R}|^2} \hat{R} \cdot |\vec{R}|$$

$$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v dv$$

$$\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} D_R \cdot R^2 \sin\theta d\theta d\phi = Q$$

$$\vec{E} = \hat{R} \frac{Q}{4\pi\epsilon R^2}$$

$$d\vec{S} = R^2 \sin\theta d\theta d\phi \hat{R}$$

$$\vec{D} \cdot d\vec{S} = \hat{R} D_R \cdot R^2 \sin\theta d\theta d\phi \hat{R}$$

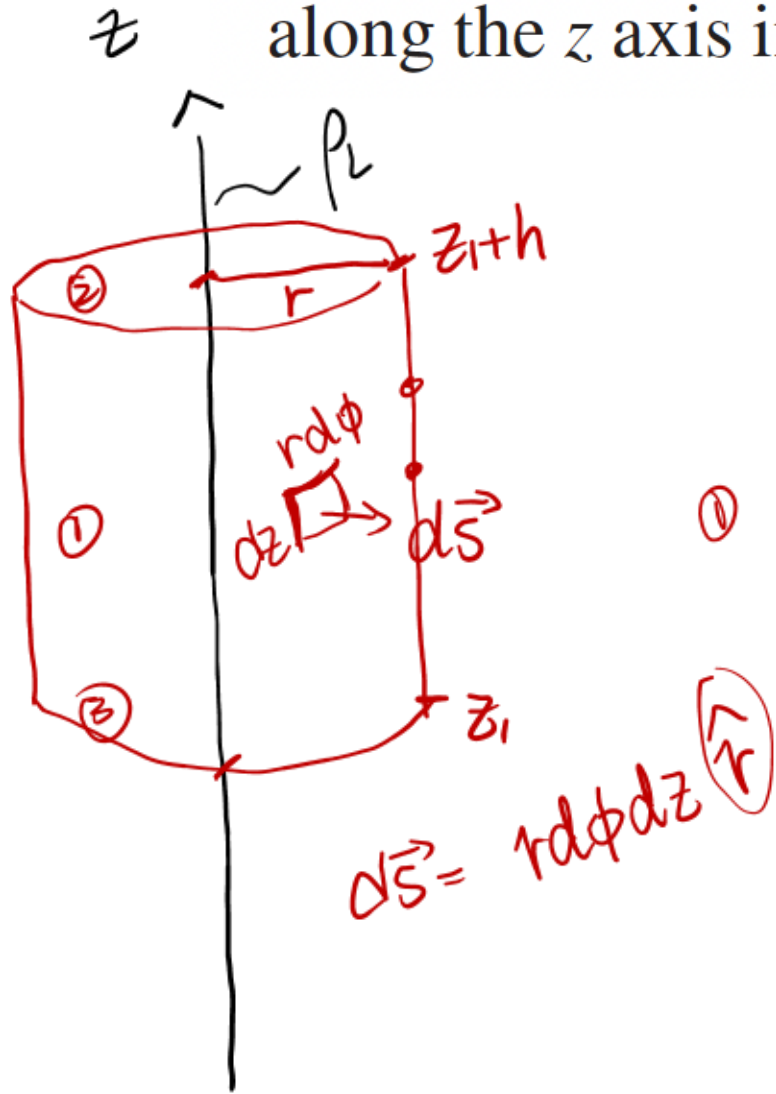
$$D_R \cdot R^2 \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin\theta d\theta$$

$$4\pi R^2 D_R = Q$$

$$D_R = \frac{Q}{4\pi R^2}$$

$$E_R = \frac{D_R}{\epsilon} = \frac{Q}{4\pi\epsilon R^2}$$

Use Gauss's law to obtain an expression for $\mathbf{E}$ due to an infinitely long line with uniform charge density ρ_ℓ that resides along the z axis in free space.

 ϵ_0

$$\vec{D} = \hat{r} D_r$$

$$\oint_S \vec{D} \cdot d\vec{S}$$

$$=$$

$$\int_V \rho_v dv$$

$$\Rightarrow \oint_S D_r \cdot r \cdot d\phi dz$$

$$\Rightarrow D_r r \cdot \int_{\phi=0}^{2\pi} d\phi \int_{z=z_1}^{z_1+h} dz$$

$$\rho_\ell = \frac{dq}{dL}$$

$$d\vec{S} = r d\phi dz \hat{r}$$

$$D_r r \cdot 2\pi \cdot h =$$

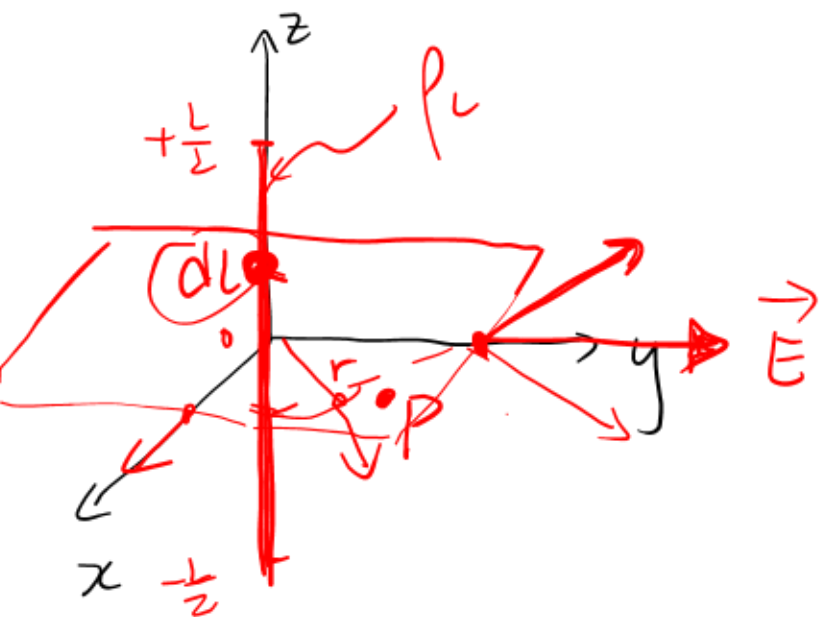
$$D_r = \frac{\rho_\ell}{2\pi r}$$

$$E_r = \frac{\rho_\ell}{2\pi r \epsilon}$$

$$\rho_\ell \cdot h \left(\frac{1}{r} \right)$$

$$\vec{E} = \hat{r} \frac{\rho_\ell}{2\pi r \epsilon}$$

Some part of the in-class problem



cylindrical
 $\hat{r}$

$$\vec{R} = \hat{r} r + \hat{z} z$$

$$f(z) = z$$



$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$

$$dQ = \rho_L dL$$

$$= \rho_L dz$$

$$d\vec{E} = \frac{1}{4\pi\epsilon} \frac{\rho_L dL}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$

$$d\vec{E} = \frac{1}{4\pi\epsilon} \frac{\rho_L dz}{(r^2 + z^2)^{\frac{3}{2}}} (\hat{r} r - \hat{z} z)$$

$$\vec{E} = \int_{z=-\frac{L}{2}}^{\frac{L}{2}} d\vec{E}$$

$$\int_{z=-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{4\pi\epsilon} \rho_L dz \cdot \frac{\hat{z} z}{(r^2 + z^2)^{\frac{3}{2}}}$$

$$\vec{R} = \hat{r} r$$

$$\vec{R}_0 = \hat{z} z$$

$$\vec{R} - \vec{R}_0 = \hat{r} r - \hat{z} z$$

$$|\vec{R} - \vec{R}_0| = \sqrt{r^2 + z^2}$$